Component Characteristics

19/10/2025

Keywords: Light-dependent resistor, variable resistor, current, potential difference, ohmic, filament, light-emitting diode

<u>Starter</u>: Complete these quiz questions about last lesson.

Quiz:

1. Volts is the unit for which measure?

2. What is the unit for resistance?

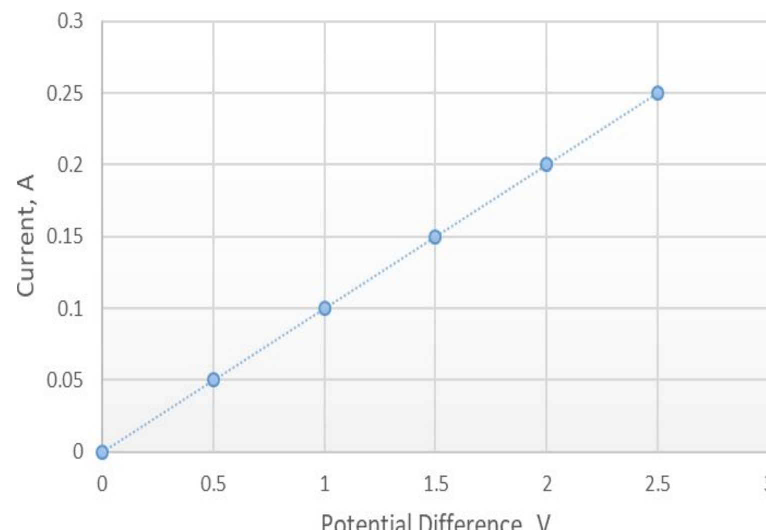
3. What equation links potential difference, current and resistance?

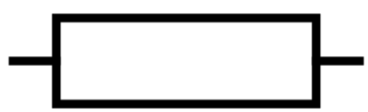
4. Potential difference is the work done by the c_________ per coulomb of _________ passing through.

5. Potential difference= E______ transferred ÷ charge

6. Resistance is a measure of how a component reduces the potential difference | current flow through a circuit.

7. Describe the trend of this graph. D__ P__



8. What is this component? 

9. Draw the circuit symbol for a cell.

10. What is the unit for charge

a) amps b) volts c) coulombs d) joules

Component Characteristics

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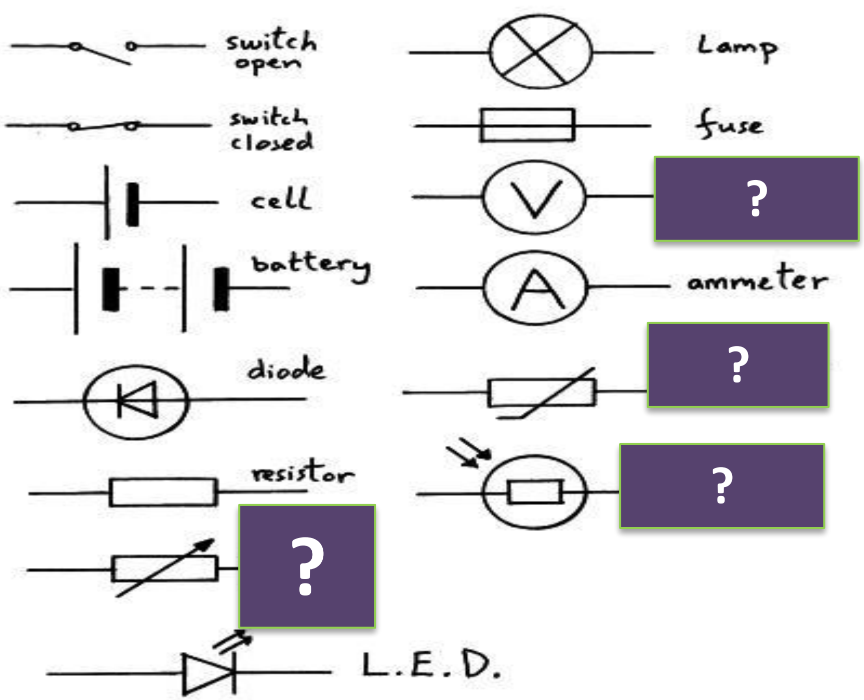
Keywords: Light-dependent resistor, variable resistor, current, potential difference, ohmic, filament, light-emitting diode

Starter: Answer these three review questions below.

1. Name the missing component names below[2]

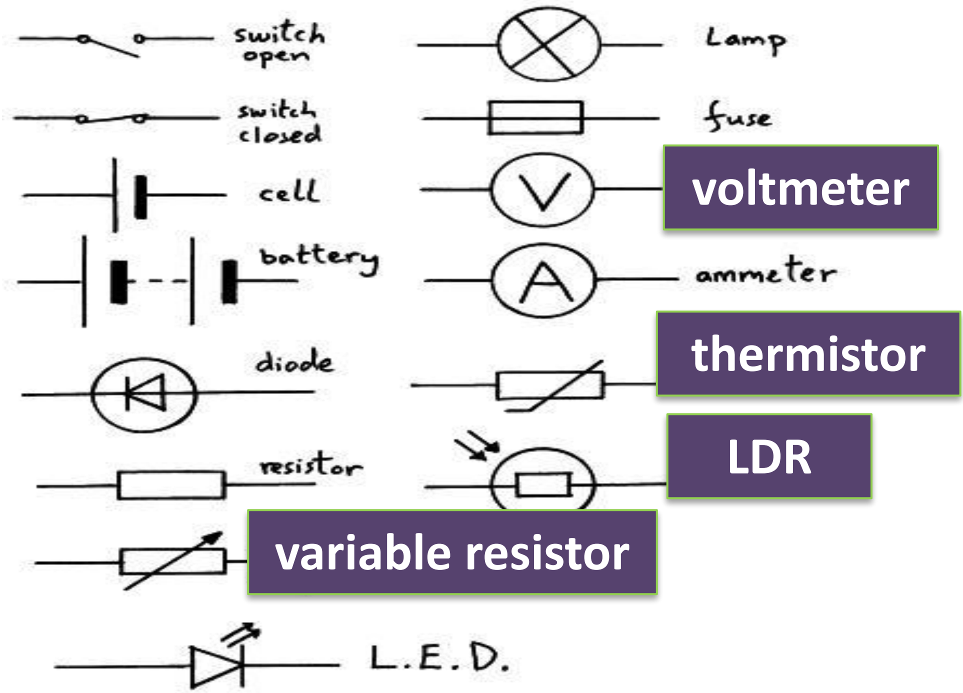


2. Ohm's law states that the _____ _______ is directly ______ to the c_____ flowing as long as the temperature remains _______ [4]

3. Calculate the resistance of the torch if the current was 0.015A and the potential difference is 12 V [3]

Review

1. Name the missing component names below[2]



2. Ohm's law states that the potential difference is directly proportional to the current flowing as long as the temperature remains constant [4]

3. Calculate the resistance of the torch if the current was 0.015A and the potential difference is 12 V [3]

$$R = \frac{V}{I} \qquad R = \frac{12\ V}{0.015\ A}$$

$$R = \mathbf{800\ \Omega}$$

Component Characteristics

19/10/2025

Keywords: Light-dependent resistor, variable resistor, current, potential difference, ohmic, filament, light-emitting diode

Create this table to review last lessons learning

Electrical property □	Resistance		
Symbol			I
Unit			$Amperes\ (A)$
Equation		$V = I\ x\ R$	

CONSOLIDATION

Electrical property	Resistance	Potential Difference	Current
Symbol	R	V	I
Unit	$Ohms\ (\Omega)$	$Volts\ (V)$	$Amperes\ (A)$
Equation	$R = \frac{V}{I}$	$V = I\ x\ R$	$I = \frac{V}{R}$

LOb: Understand current-voltage characteristics of different components in a circuit.

Keywords:

Light-dependent resistor, variable resistor, current, potential difference, ohmic, filament, light-emitting diode

Success Criteria:

- Describe the current voltage characteristic of a filament bulb
- Describe the voltage characteristic of a diode
- Describe the current-voltage characteristic of thermistors and LDRs

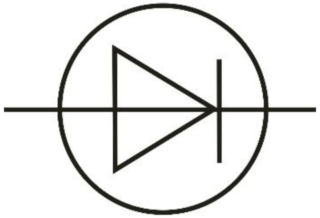
Did you know… this man called Andre Marie Ampere was a 19th century French physicist from which we get the unit for current.

This lesson is about investigating the current and potential difference of different circuit components.

Filament bulb

**Resist
or**

Dio



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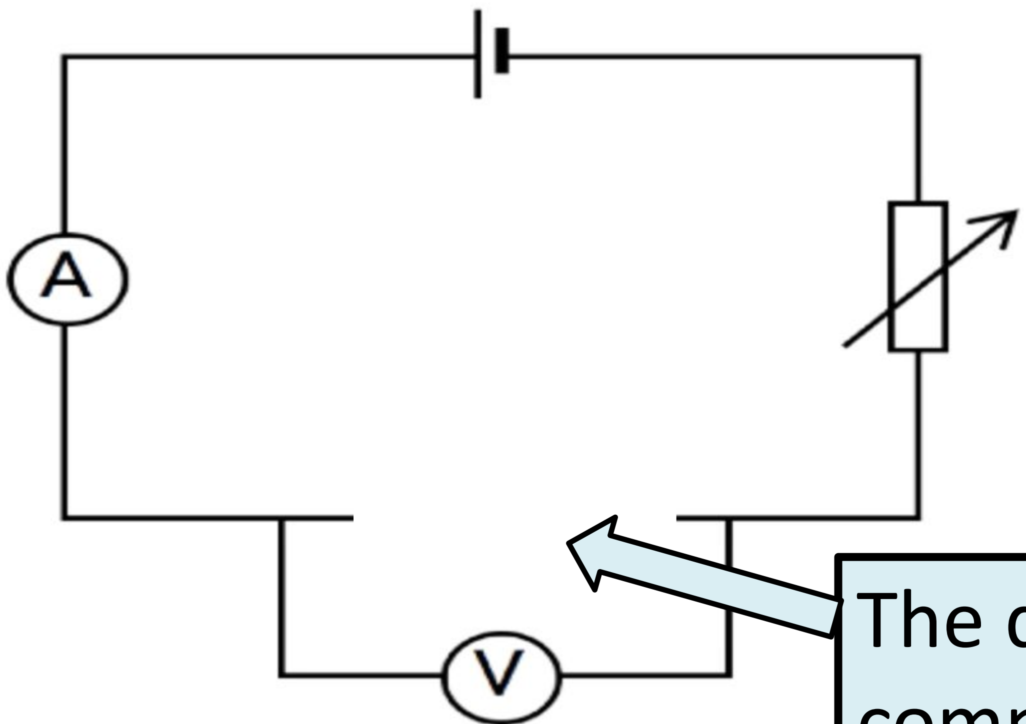
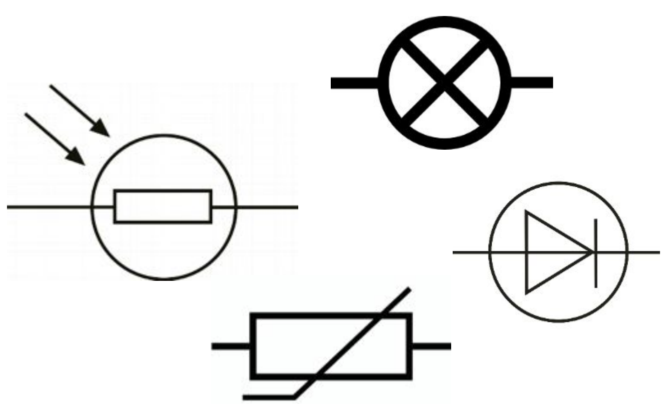
ACTIVATION

Each component will be tested in a circuit that looks like this…

Required Practical



The different components get added into here



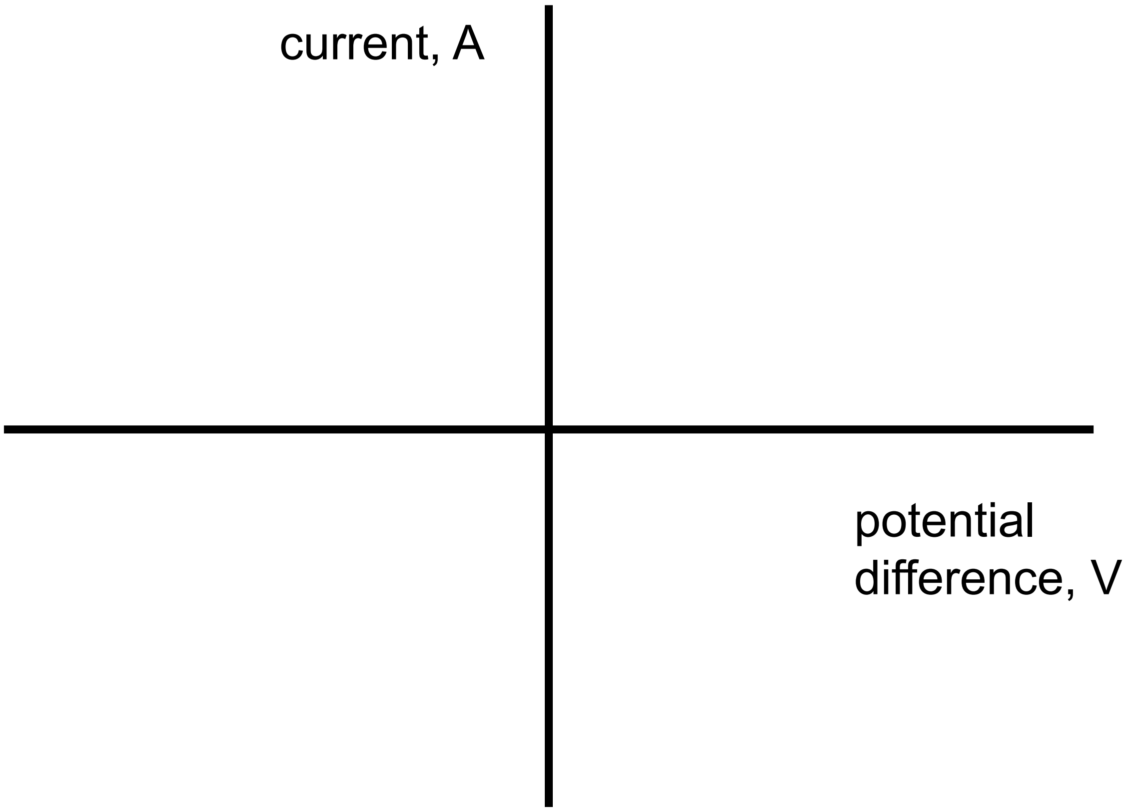
WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

POWER!

The results for each component are plotted onto a graph. The graph looks different for each component.



$$R = \frac{V}{I}$$

We can then calculate the resistance.

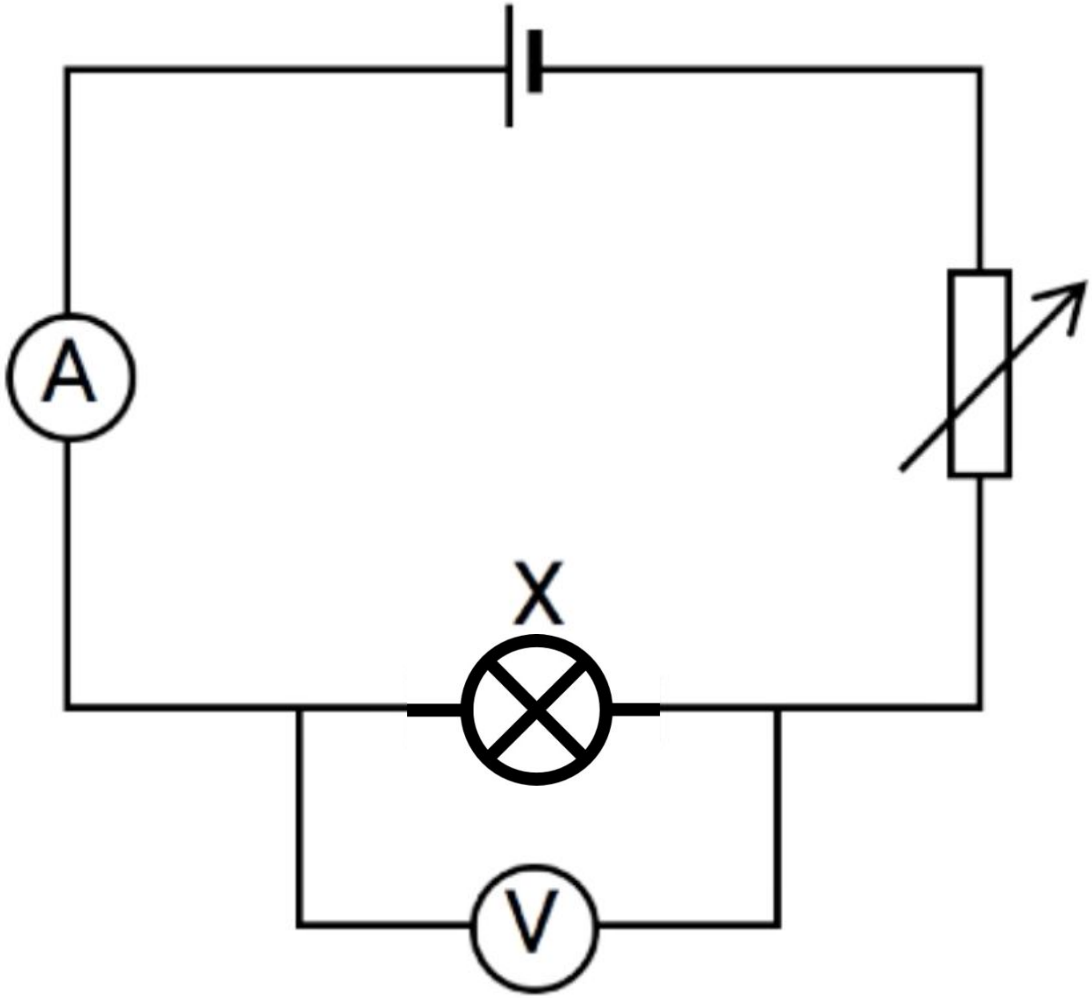
Activity: Watch the set up of the equipment

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

POWER!



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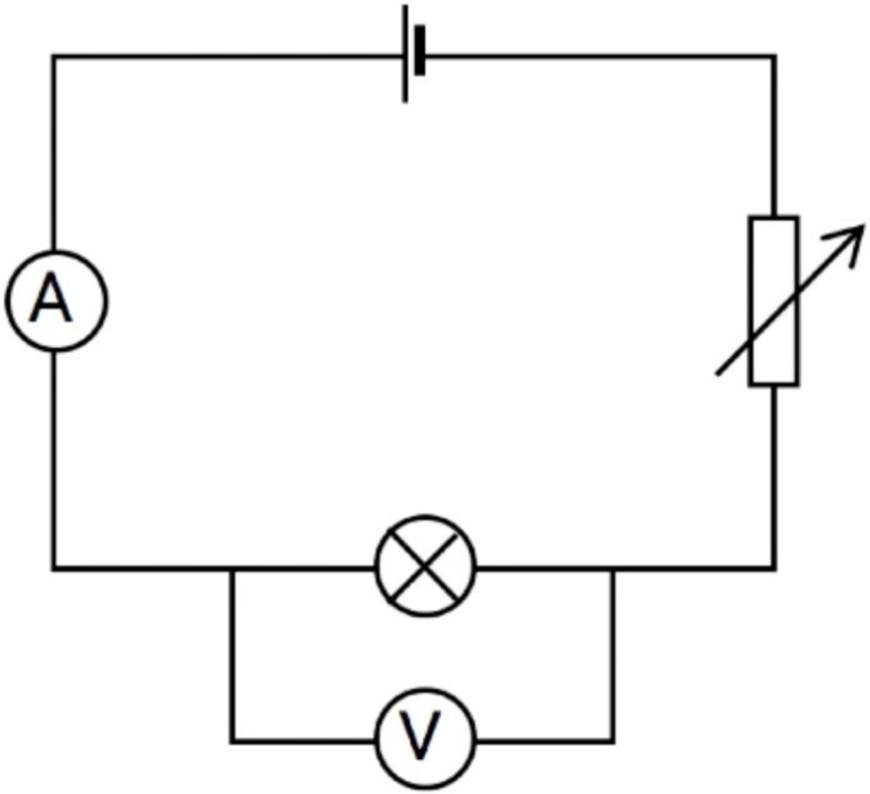
ACTIVATION

METHOD

1. Set the variable resistor to the largest resistance and record I and V through the component.

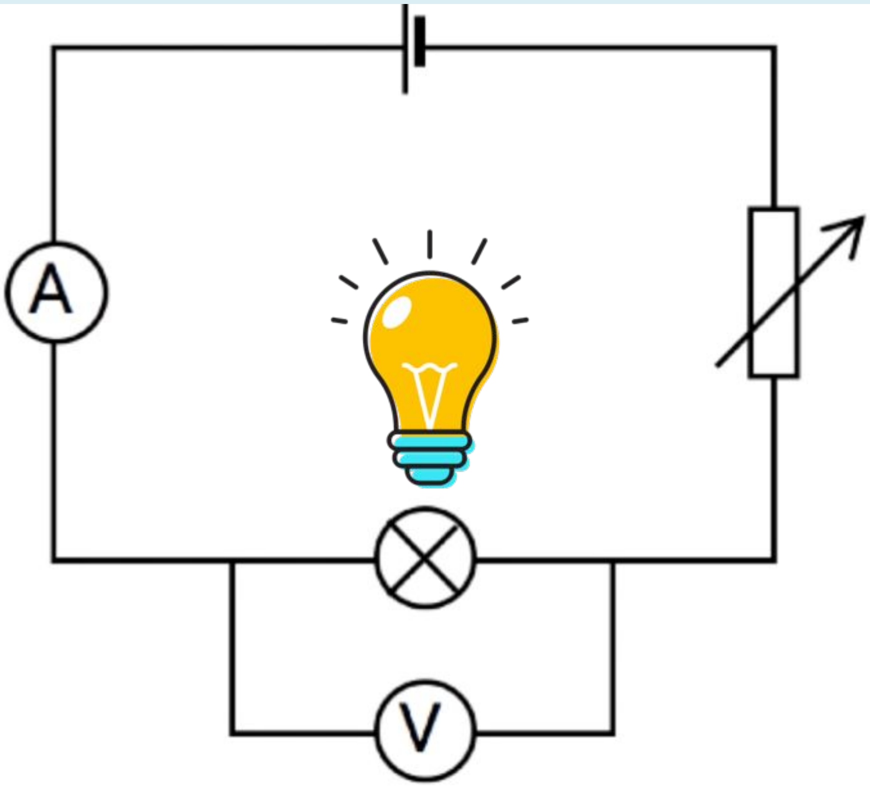


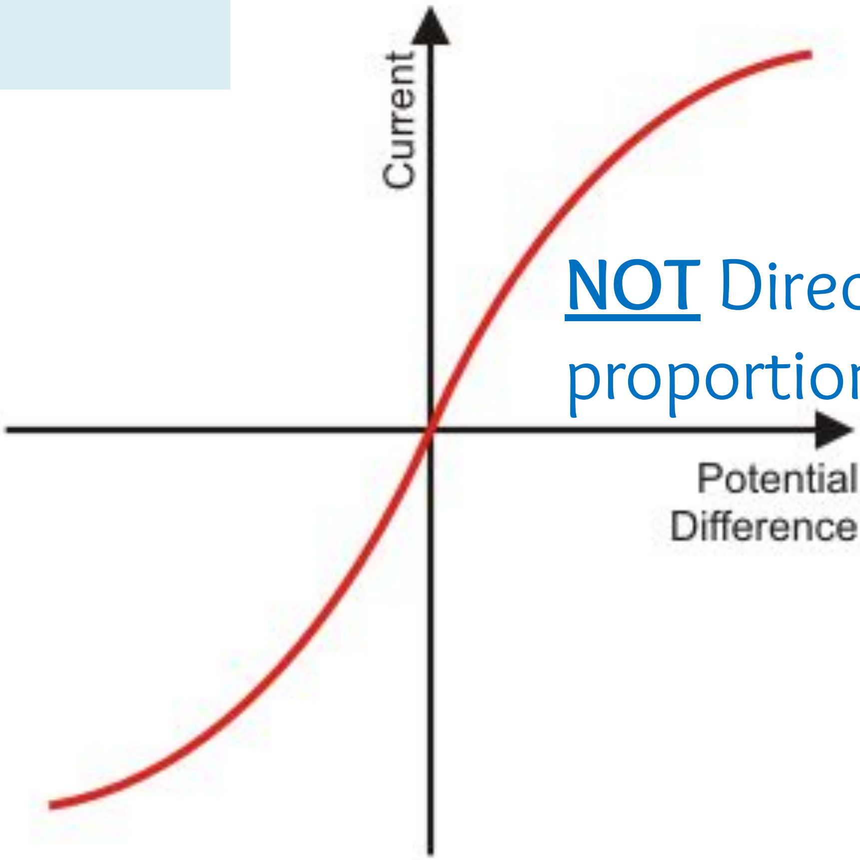
2. Set the variable resistor to 5 or 6 alternate values and repeat the measurements. Finish with the variable resistor at the lowest resistance.

This is the results for a filament lamp.

Ohm's law states that current is directly proportional to the potential difference and the resistance remains constant when temperature is constant.

Filament bulbs are **non-ohmic** components





NOT Directly proportional

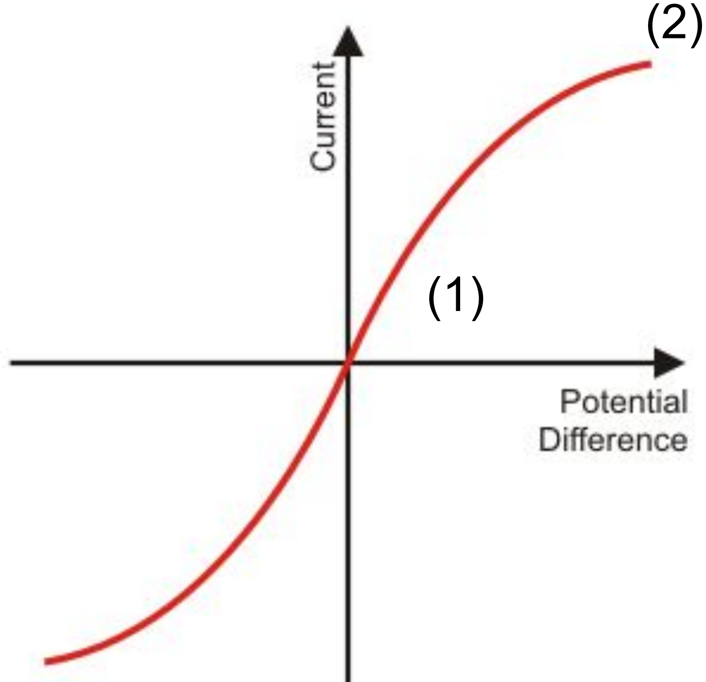
A bulb does not obey Ohm's law.

explanation 1

Low I/V values so resistance is low. A low current means there is less heat in the filament bulb. It is easy for electrons to pass through the bulb.

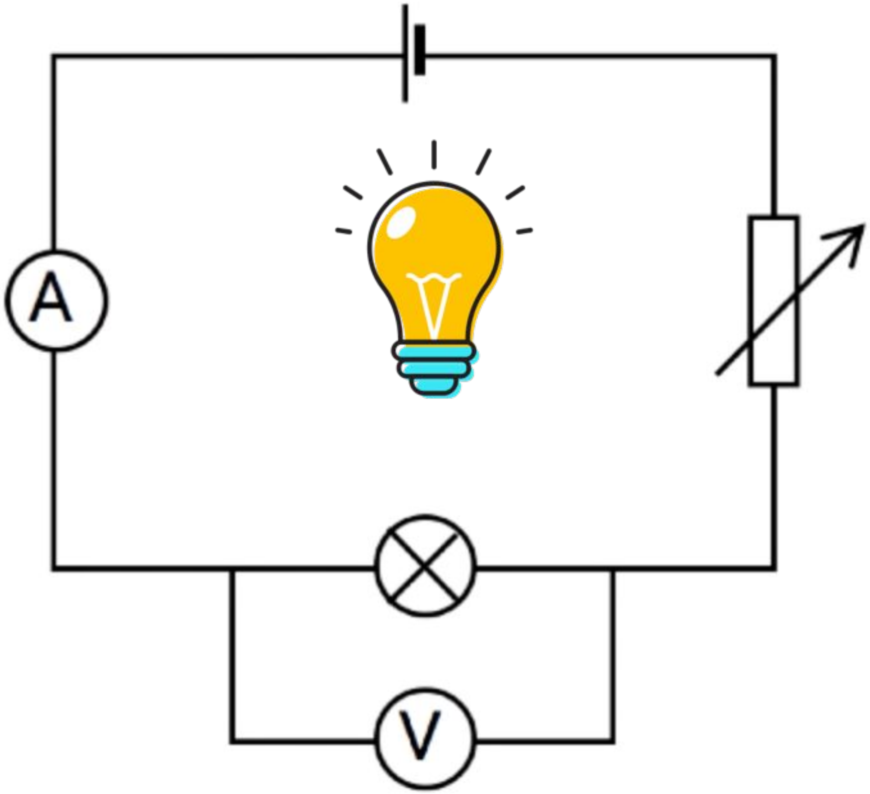
explanation 2

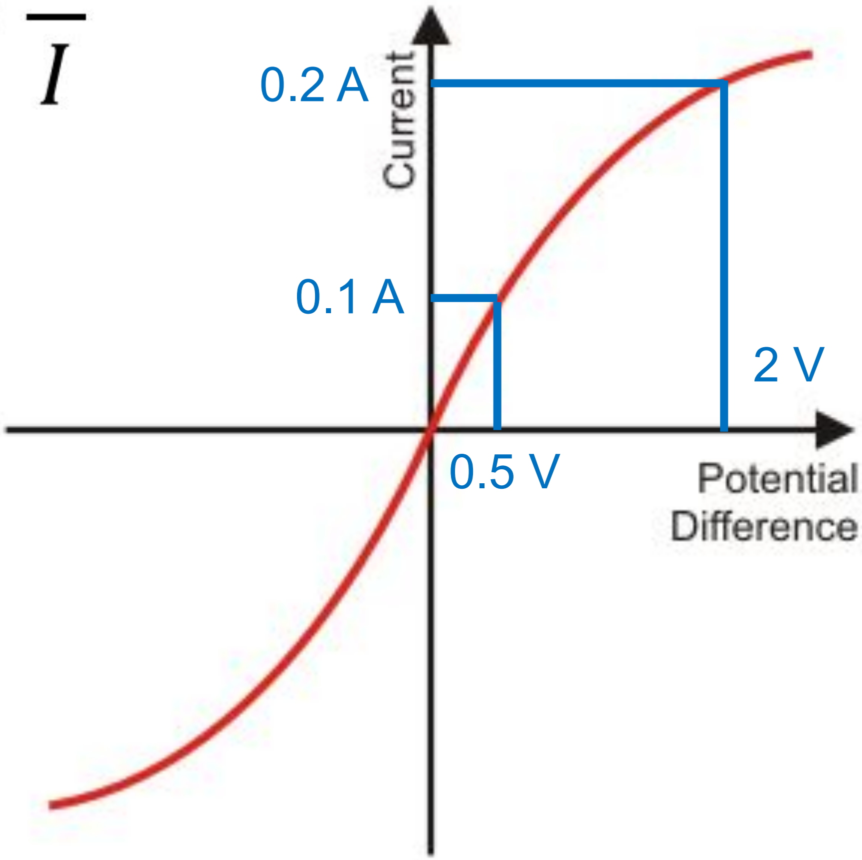
The high current increases the heat of the filament causing atoms to vibrate more. This makes it harder for the electrons to flow through. Therefore, resistance is high.



Activity: 1. Draw the circuit diagram for this practical.
2. Sketch the shape of the I-V characteristics.
3. Describe whether it obeys Ohm's law.
4. Calculate the resistance for the two points on the graph.

$$R = \frac{V}{I}$$





LOb: Understand current-voltage characteristics of different components in a circuit.

Keywords:
Light-dependent resistor, variable resistor, current, potential difference, ohmic, filament, light-emitting diode

Success Criteria:

- ✔ Describe the current voltage characteristic of a filament bulb
- Describe the voltage characteristic of a diode
- Describe the current-voltage characteristic of thermistors and LDRs

Did you know… this man called Andre Marie Ampere was a 19th century French physicist from which we get the unit for current.

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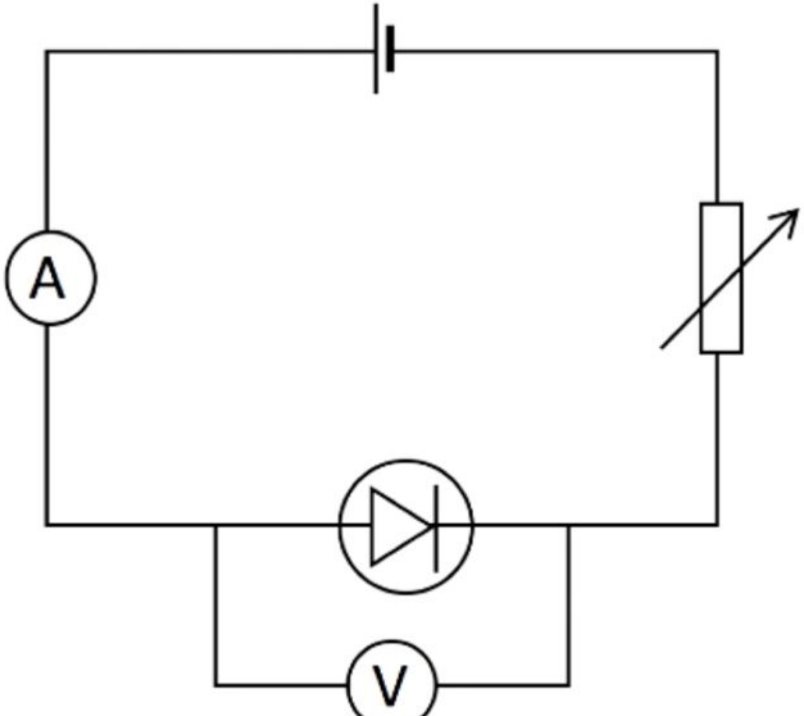
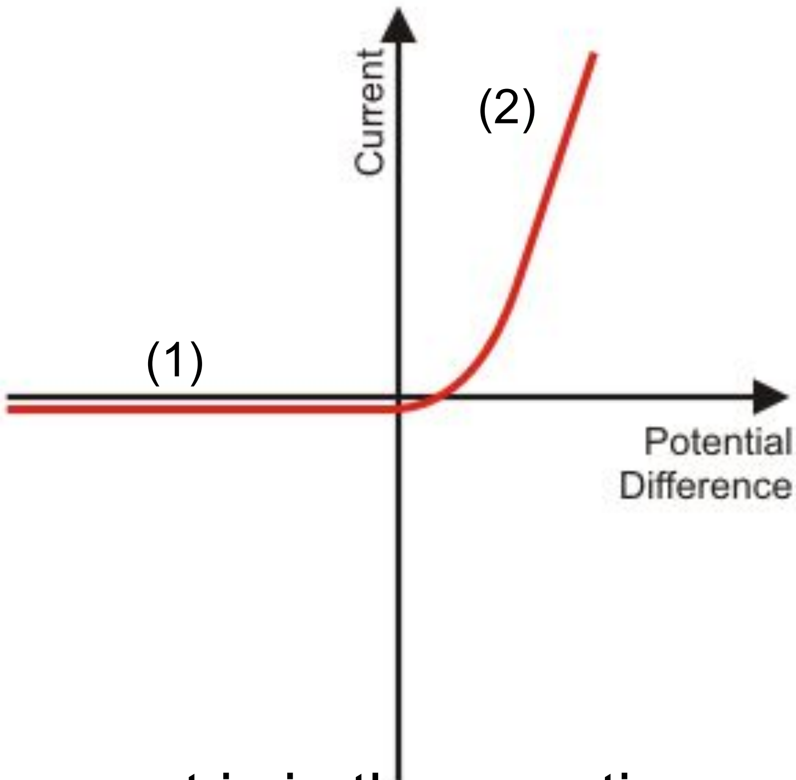
CONNECTION

Here is the shape of a diode: Diodes allow current to flow in one direction.





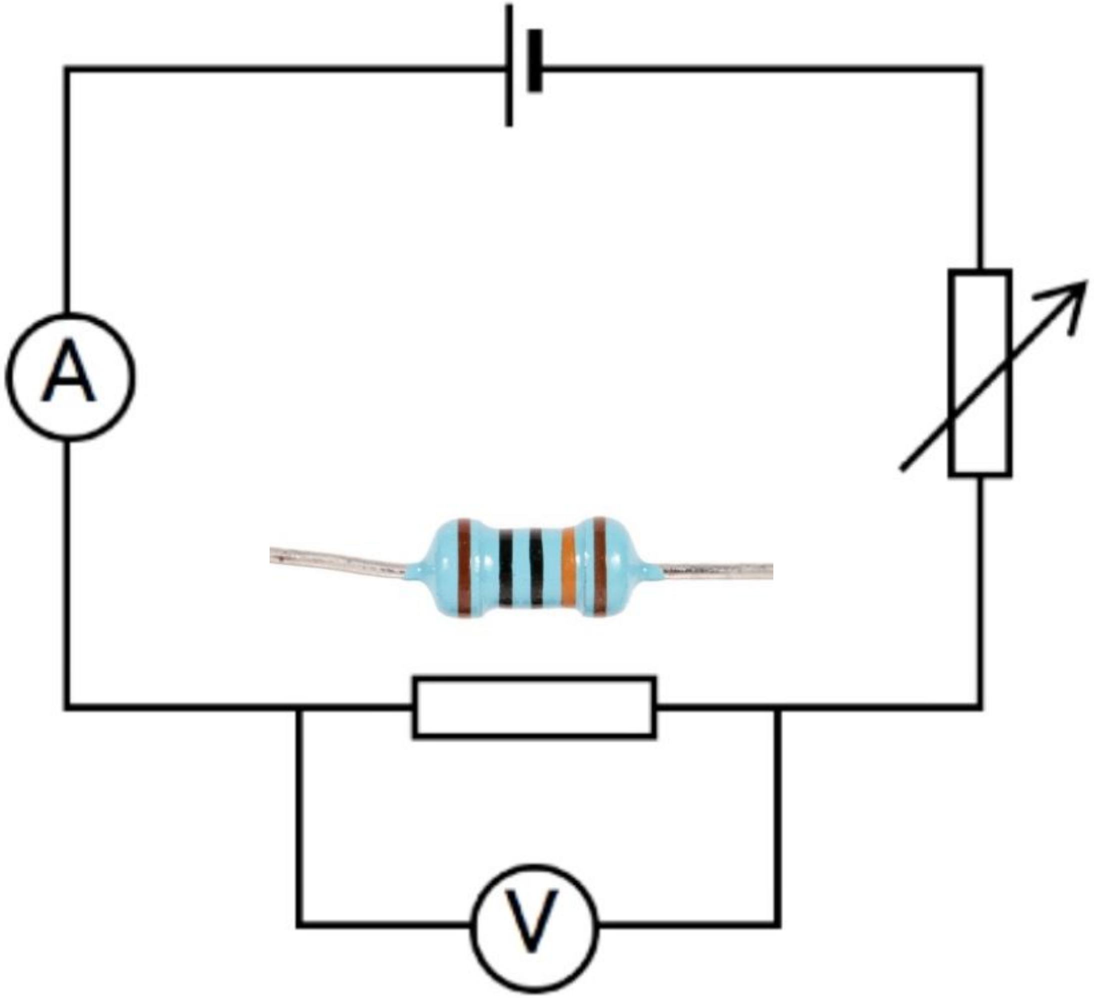
1 explanation

At this point the current is in the negative direction the diode doesn't let current flow in this direction. Resistance is high.

2 explanation

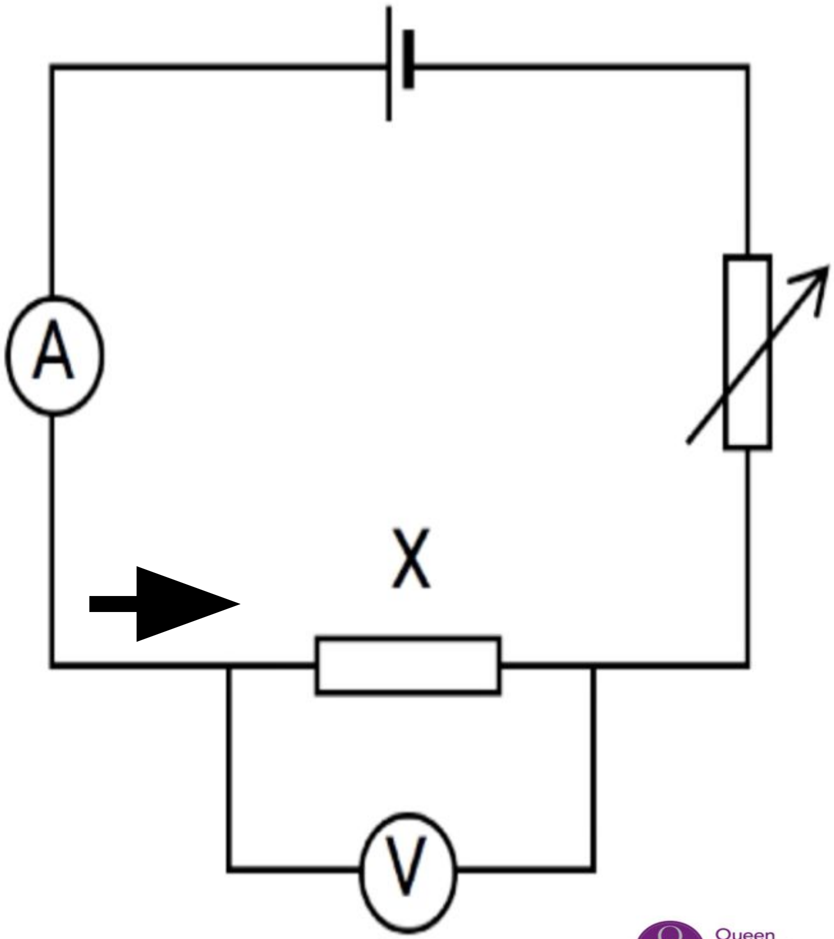
Current is not directly proportional to p.d. Resistance changes as current changes.

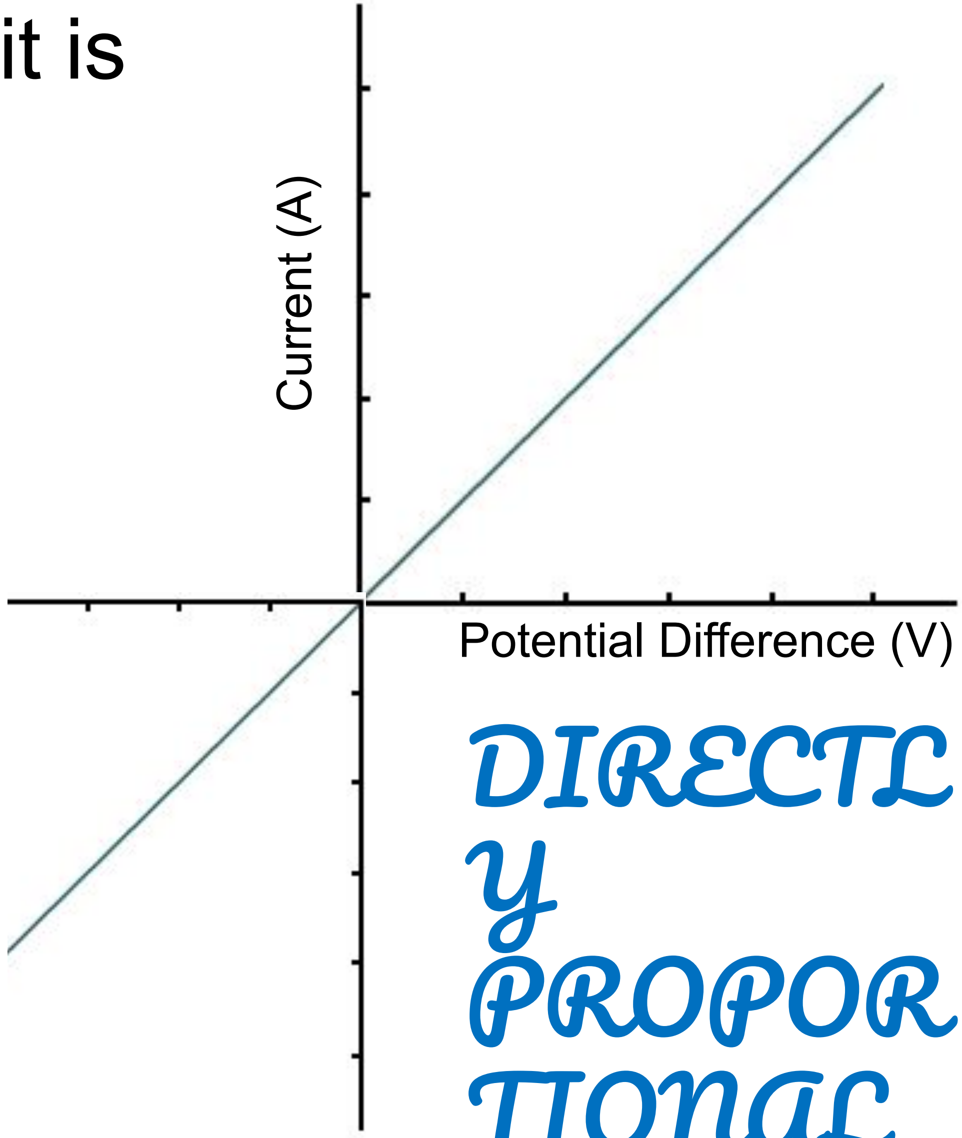
The final test is with a resistor



This is the graph for a resistor

It obeys ohm's law so it is an ohmic conductor.

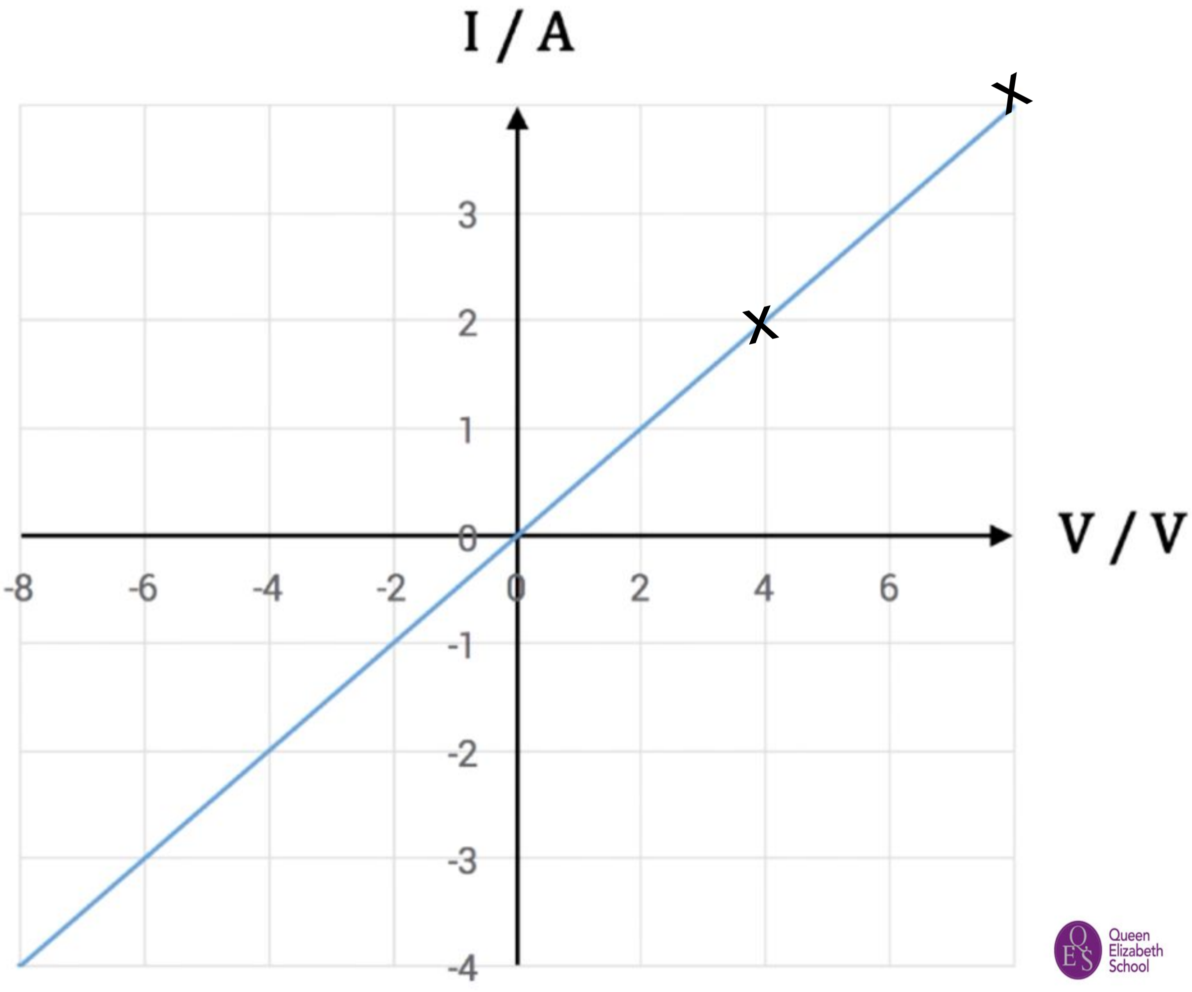




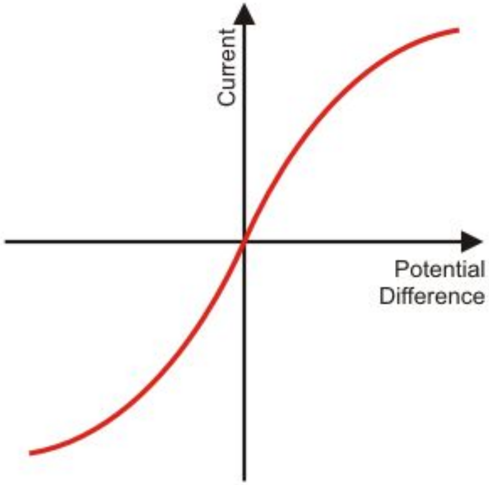
DIRECTLY PROPORTIONAL

Lets prove it

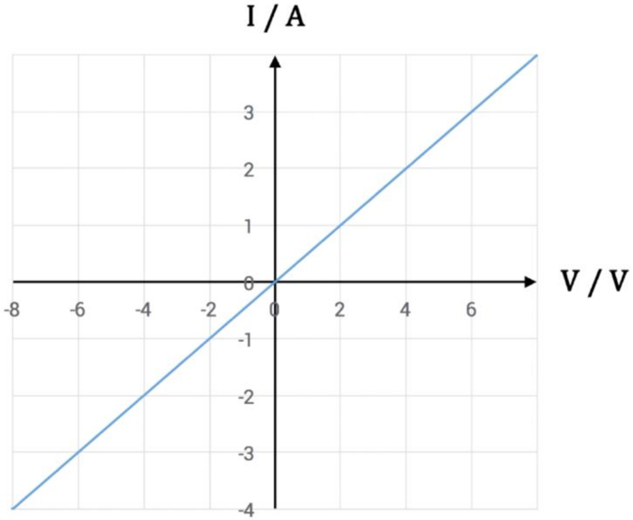
Calculate the resistance of the two points on the graph.



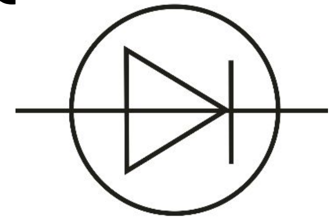
Filament bulb

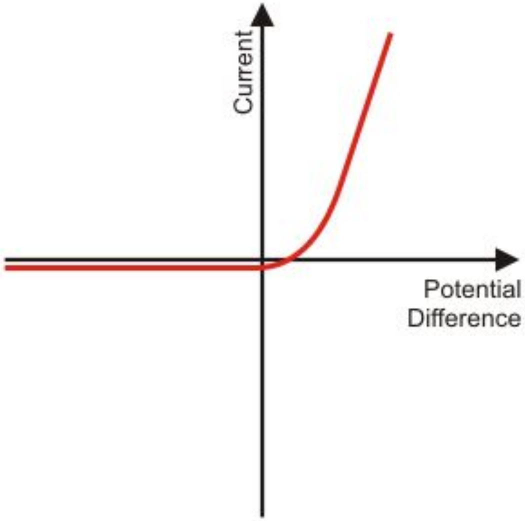


Resist
or



Dio





Lets do a mini white board review

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CONSOLIDATION

LOb: Understand current-voltage characteristics of different components in a circuit.

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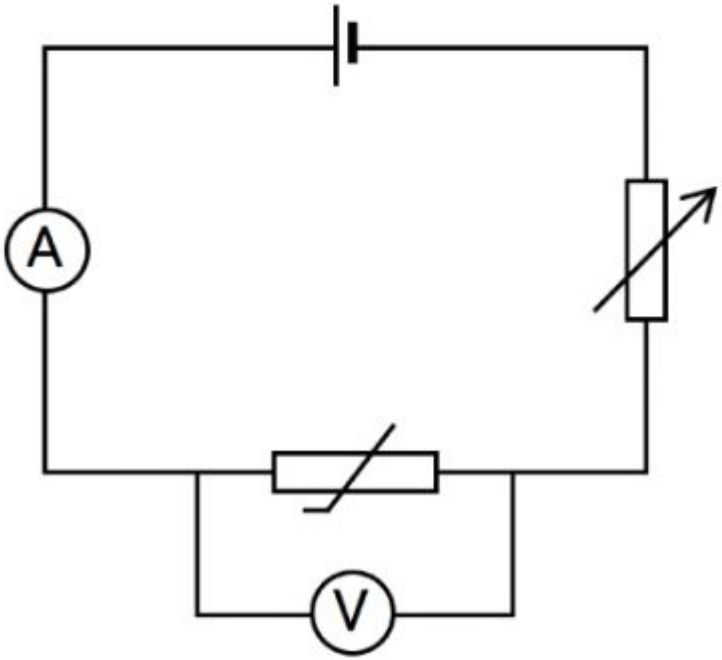
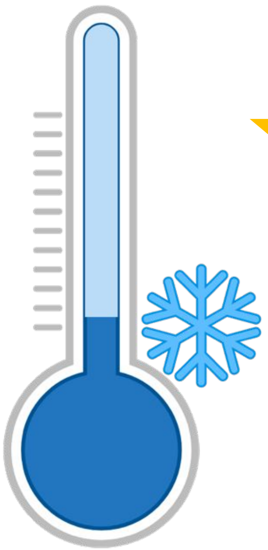
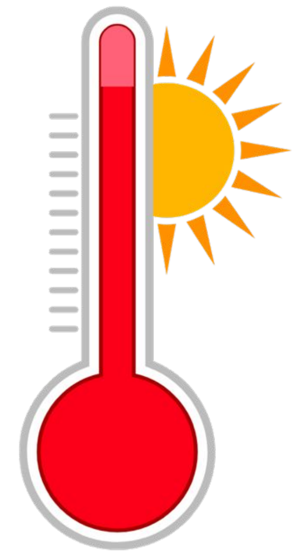
CONNECTION

Thermistors change resistance based on the temperature.

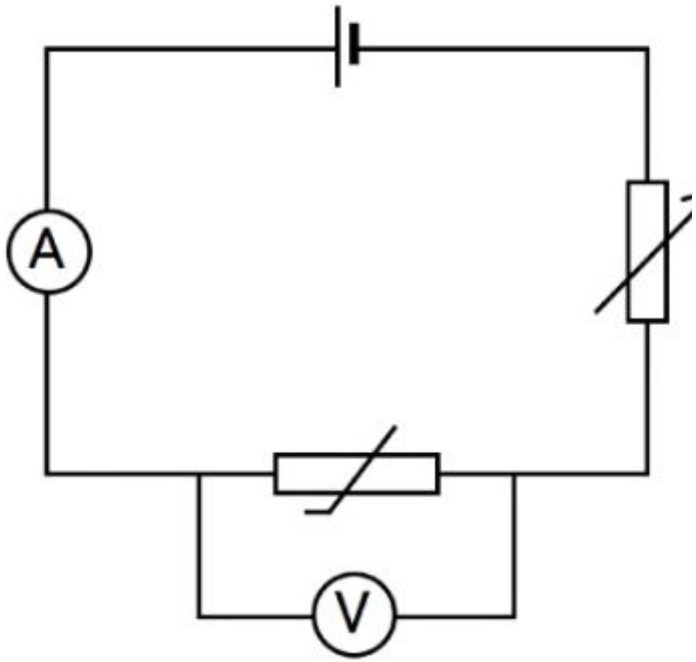




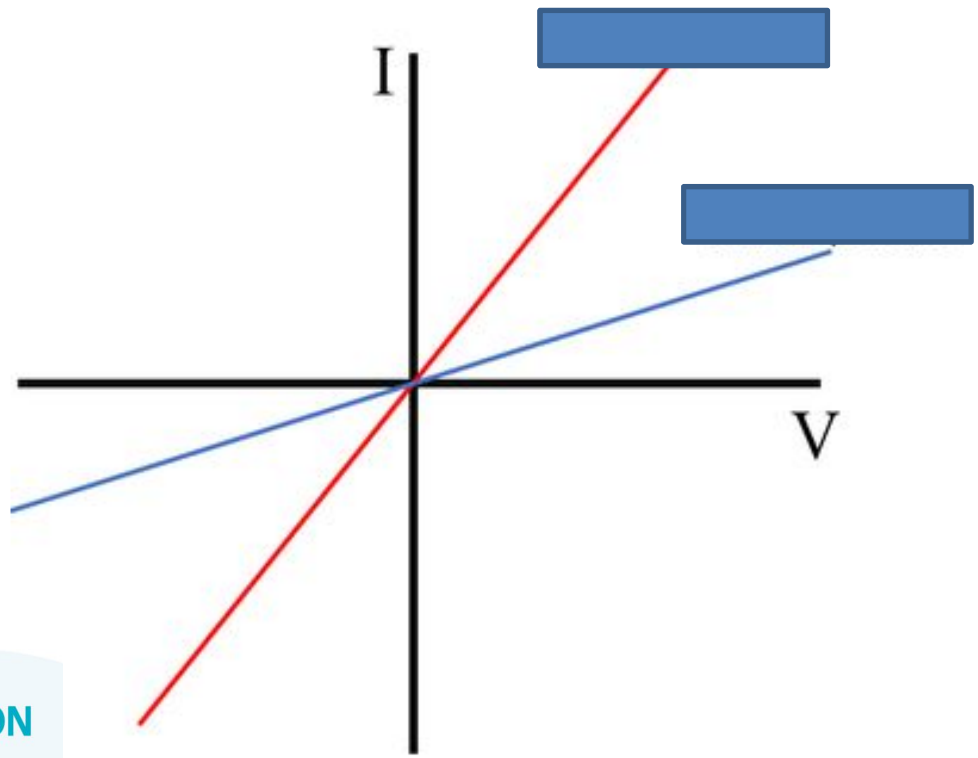
Resistance (Ω) decreases as temperature increases



Thermistors: Resistance decreases as temperature increases.



An engineer was investigating the behaviour of a thermistor at different temperatures. Which of the below graphs was obtained at 25°C, and which one was obtained at 100°C?



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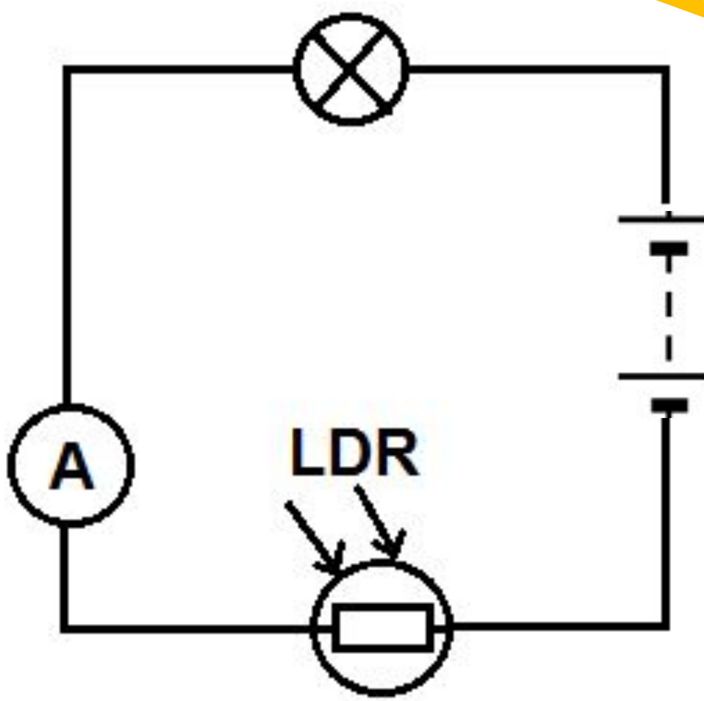
ACTIVATION

Light dependent resistors (LDR) change resistance based on the light intensity.

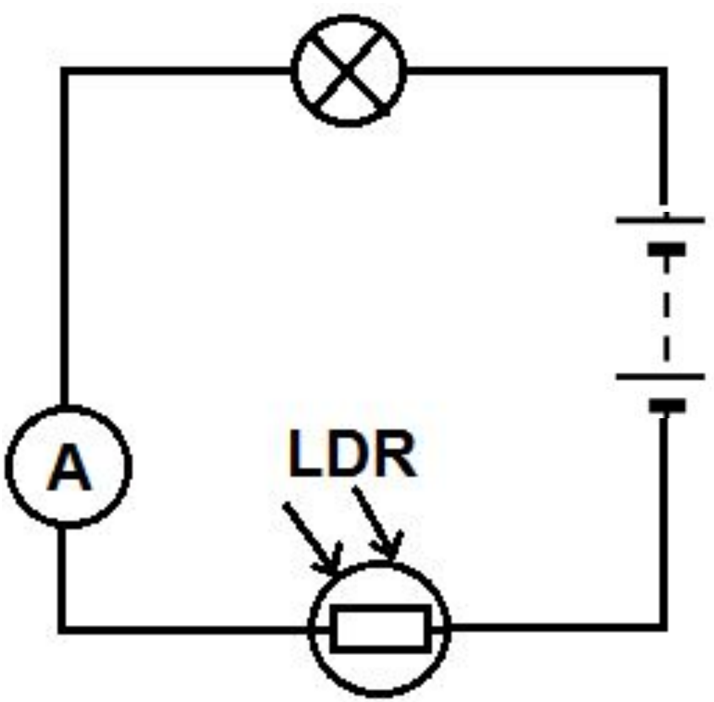
Resistance (Ω) decreases as light intensity increases

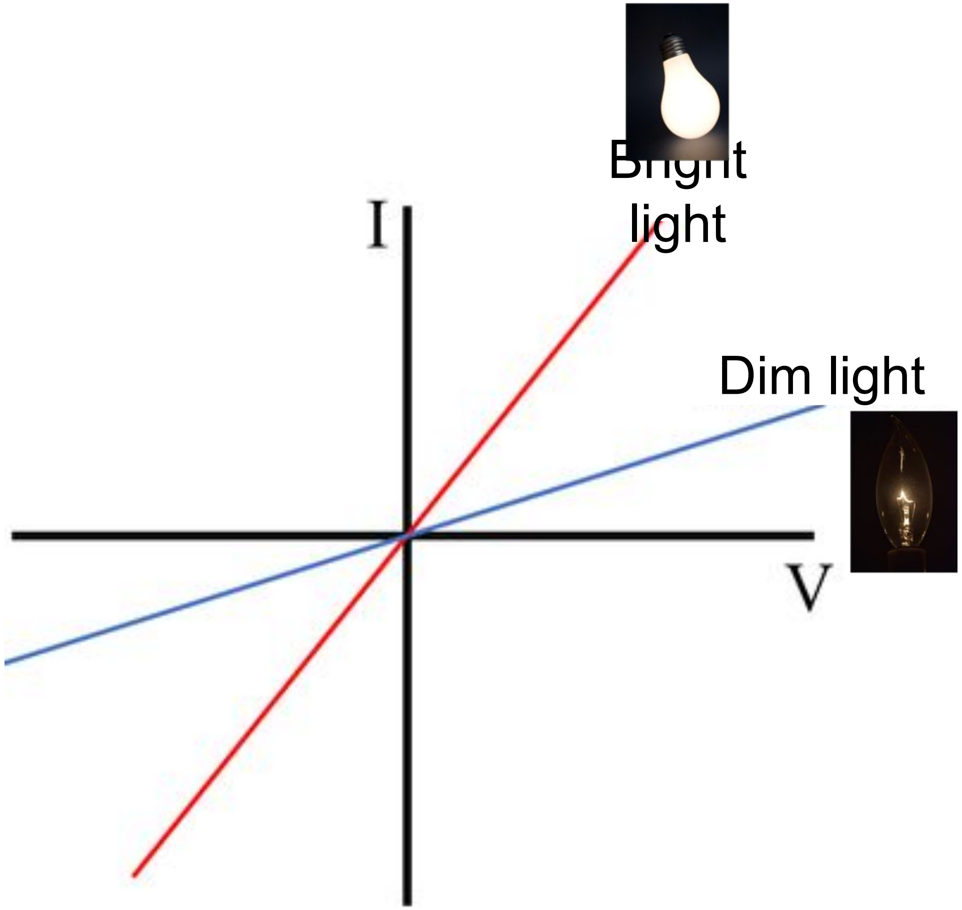
Di
m
Lig
ht



Brig
ht
Light

LDRs: Resistance decreases as light intensity increases.





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ACTIVATION

Activity: Chose the correct word for each sentence

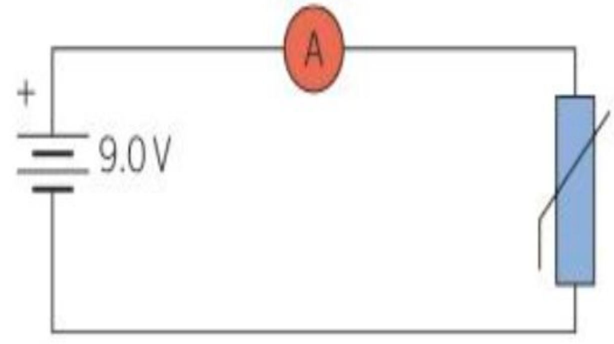
LDRs are a type of ***resistor* | *diode* | *resistance*** that changes the ***light* | *resistance*** of a circuit depending on the light intensity. If the light intensity ***increases* | *decreases*** the resistance ***increases* | *decreases.*** Thermistors change resistance based on ***light intensity* | *current* | *temperature.*** If the temperature increases the resistance ***increases* | *decreases.***

Activity: Chose the correct word for each sentence

LDRs are a type of ***resistor* | *diode* | *resistance*** that changes the ***light* | *resistance*** of a circuit depending on the light intensity. If the light intensity ***increases* | *decreases*** the resistance ***increases* | *decreases*.** Thermistors change resistance based on ***light intensity* | *current* | *temperature*.** If the temperature increases the resistance ***increases* | *decreases*.**

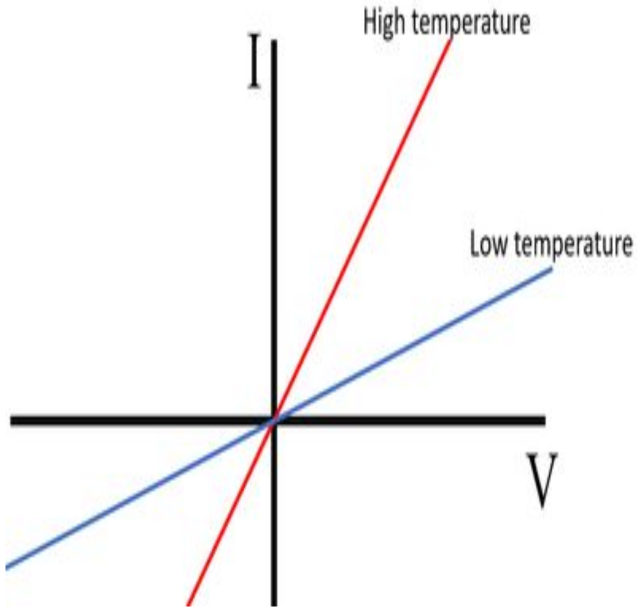
Activity: Complete these questions about

1. What is the blue circuit symbol called?



2. Which way is current flowing in the circuit. clockwise | anti-clockwise

3. At 15°C the current through the ammeter is 0.6 A and the potential difference is 9.0 V. Calculate the resistance of the circuit.

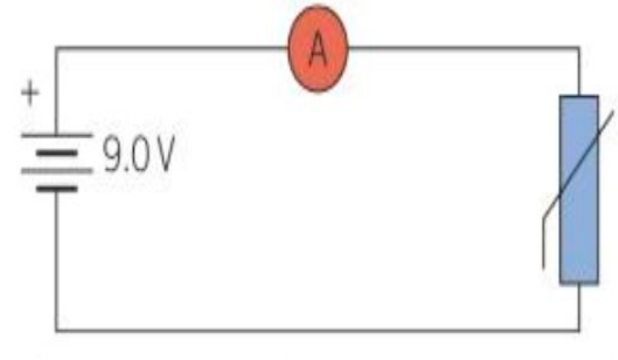


4. If the temperature is increased to 30°C what happens to the resistance.

5. Using the graph, what happens to the current if the temperature increases?

6. Name another device that changes resistance based on an environmental factor.

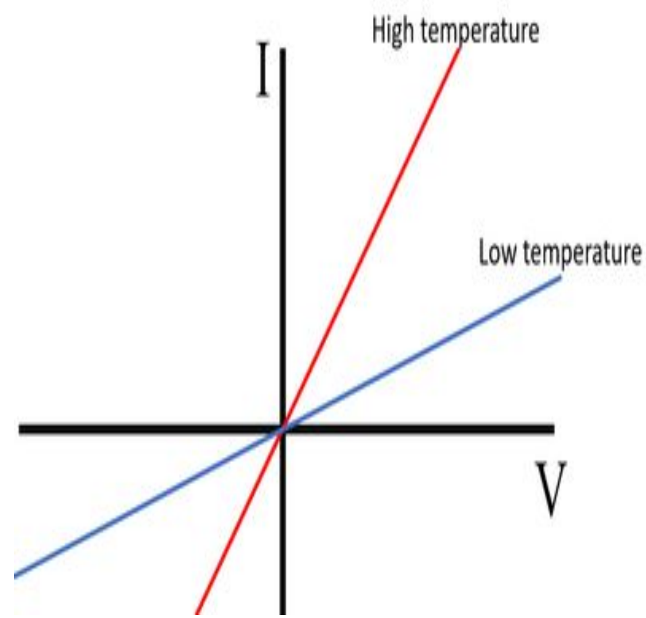
CONSOLIDATION



1. What is tl **thermistor** symbol called?

2. Which way is current flowing in the circuit. clock **Anticlockwise** ckwise

3. At 15°C the current through the ammeter is 0.6 A and the potentia **Resistance = 9 ÷ 0.6 □ R = 15** the resistance of the circuit.
Ω



4. If the temperature is increased **Resistance will decrease** to 30°C what happens to the resistance.

5. Using the graph, what happens to **The current increases** the current if the temperature increases?

6. Name another device that changes **Light dependent resistor** resistance based on an environmental factor.

LOb: Understand current-voltage characteristics of different components in a circuit.

Keywords:

Light-dependent resistor, variable resistor, current, potential difference, ohmic, filament, light-emitting diode

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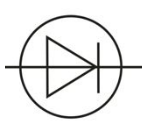
Did you know… this man called Andre Marie Ampere was a 19th century French physicist from which we get the unit for current.

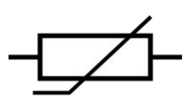
keyfacts

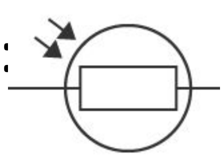
CONSOLIDATION

Resistance equation: $R = \dfrac{V}{I}$

The filament lamps  resistance increases if the filament lamp **temperature increases.**

Diode:  forward resistance low: reverse resistance high

Thermistor:  As the temperature increase the resistance decreases

LDR:  As the light intensity increase resistance decreases

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CONSOLIDATION